

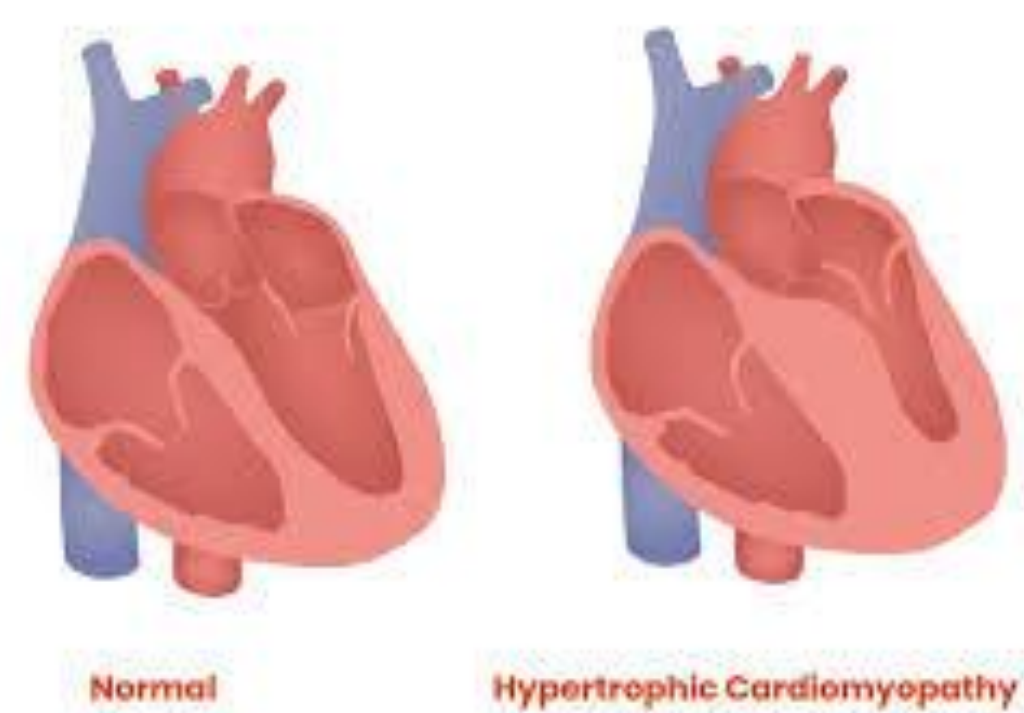
The Effects of Low-Intensity Exercise on *Drosophila melanogaster* Induced with Cardiac Hypertrophy

TMED

Background

Hypertrophic Cardiomyopathy (HCM) is a heart disease that affects 1.7 million Americans. Due to side effects that occur with HCM 1 in 4 people diagnosed with cardiac hypertrophy dies within 5 years of the diagnosis.

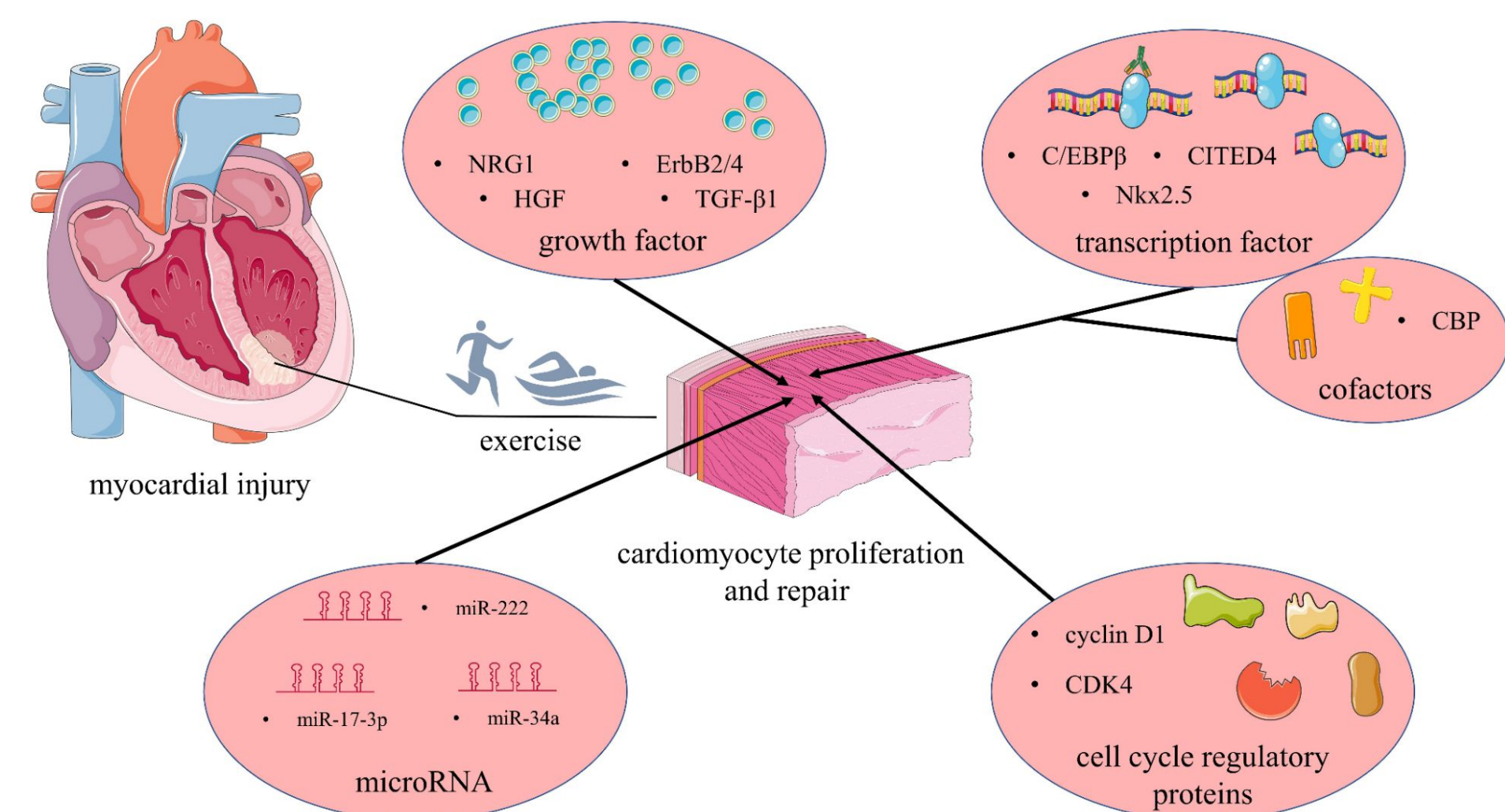
Cardiac Hypertrophy



- Thickening of the heart muscle
- Increased actin and myosin production
- Impaired relaxation
- Increased risk of heart failure and blood clots

<https://pennstatehealthnews.org/2023/08/the-medical-minute-understanding-hypertrophic-cardiomyopathy/>

Exercise



- Increased duration of exercise leads to periodic metabolic stress. Cardiomyocyte proliferation and repair is promoted, leading to decreased severity of HCM.

Figure 1: Effect of exercise on HCM. Images were taken by the researcher from the sources cited under each image.

Purpose

The purpose of this study is to determine if low-intensity exercise can reduce the severity of HCM and serve as a viable treatment option for the millions impacted by the disease.

Variables/Hypothesis

Hypothesis: Low intensity exercise will decrease the severity of HCM in *Drosophila melanogaster* model of HCM, with higher durations of exercise leading to lower levels of HCM.

IV: Exercise duration (15, 30, 60 minutes)

DV: Expression of actin in the heart and climbing assay

Methods

	Cardiac Hypertrophy	Vibrations	Exercise
Negative Control 1	No (<i>Drosophila melanogaster</i>)	No	No
Negative Control 2	Yes (<i>UAS-Ras85D^{V12}</i> x <i>Tin C-Gal4</i>)	No	No
Condition Control	Yes (<i>UAS-Ras85D^{V12}</i> x <i>Tin C-Gal4</i>)	Yes	No
Experimental Group	Yes (<i>UAS-Ras85D^{V12}</i> x <i>Tin C-Gal4</i>)	Yes	Yes

Table 1: Outline of Groups. This table was created by the researcher in Google Slides

Exercise Regimens

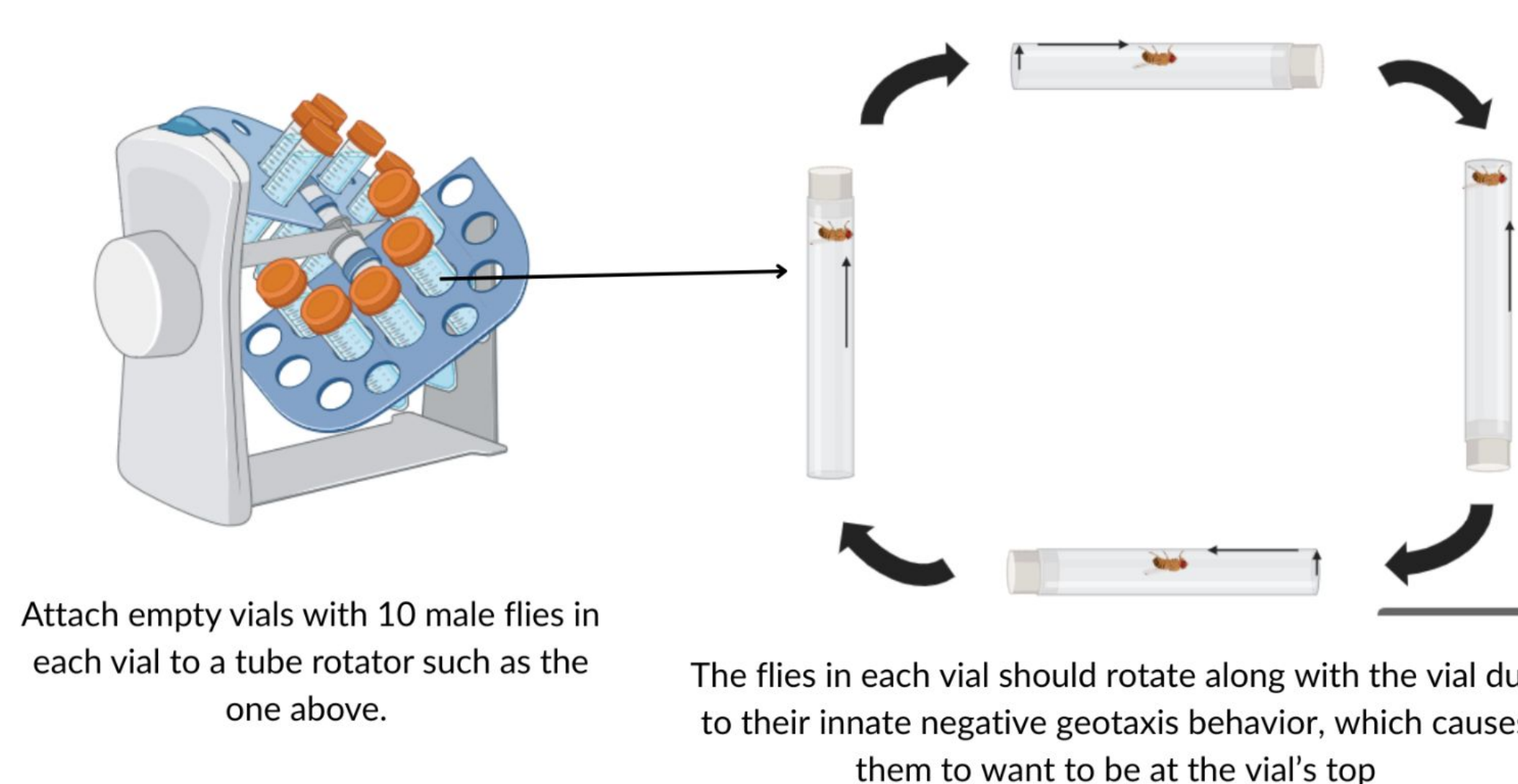


Figure 2: Graphic created by the researcher in BioRender

Climbing Assay

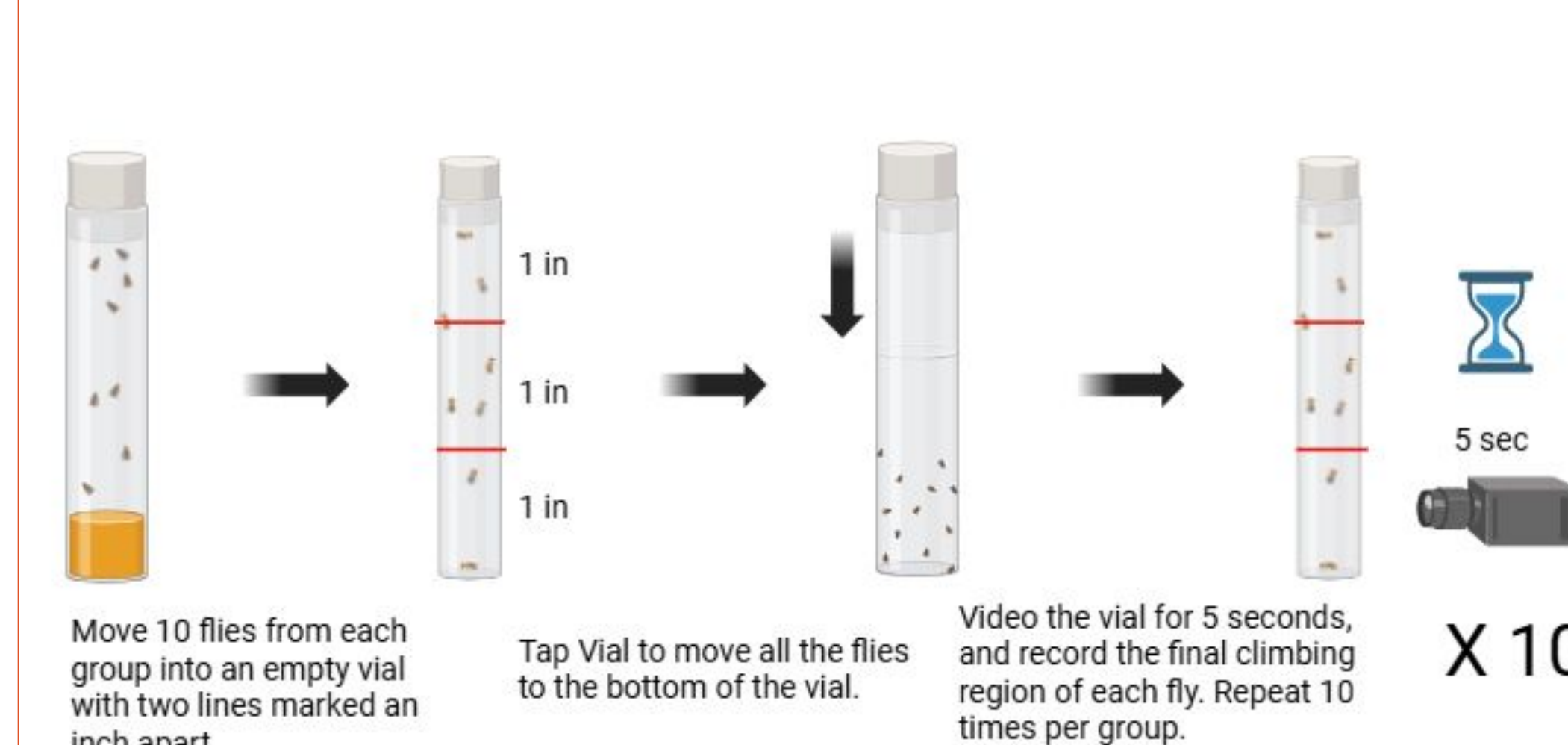


Figure 3: Graphic created by the researcher in BioRender

Western Blot Analysis

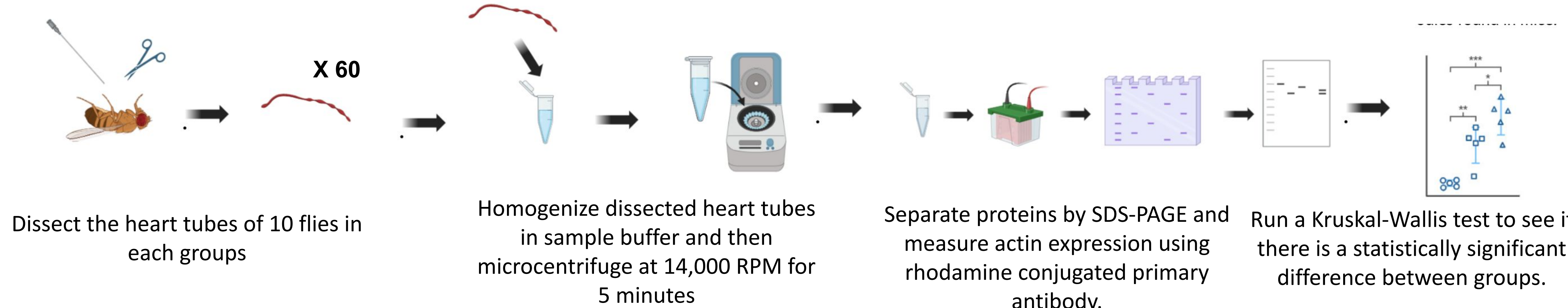


Figure 4: Cardiac hypertrophy is measured using a western blot against the protein actin. Increased expression of actin correlates with increased severity of cardiac hypertrophy. Graphic created by researcher in BioRender

Results

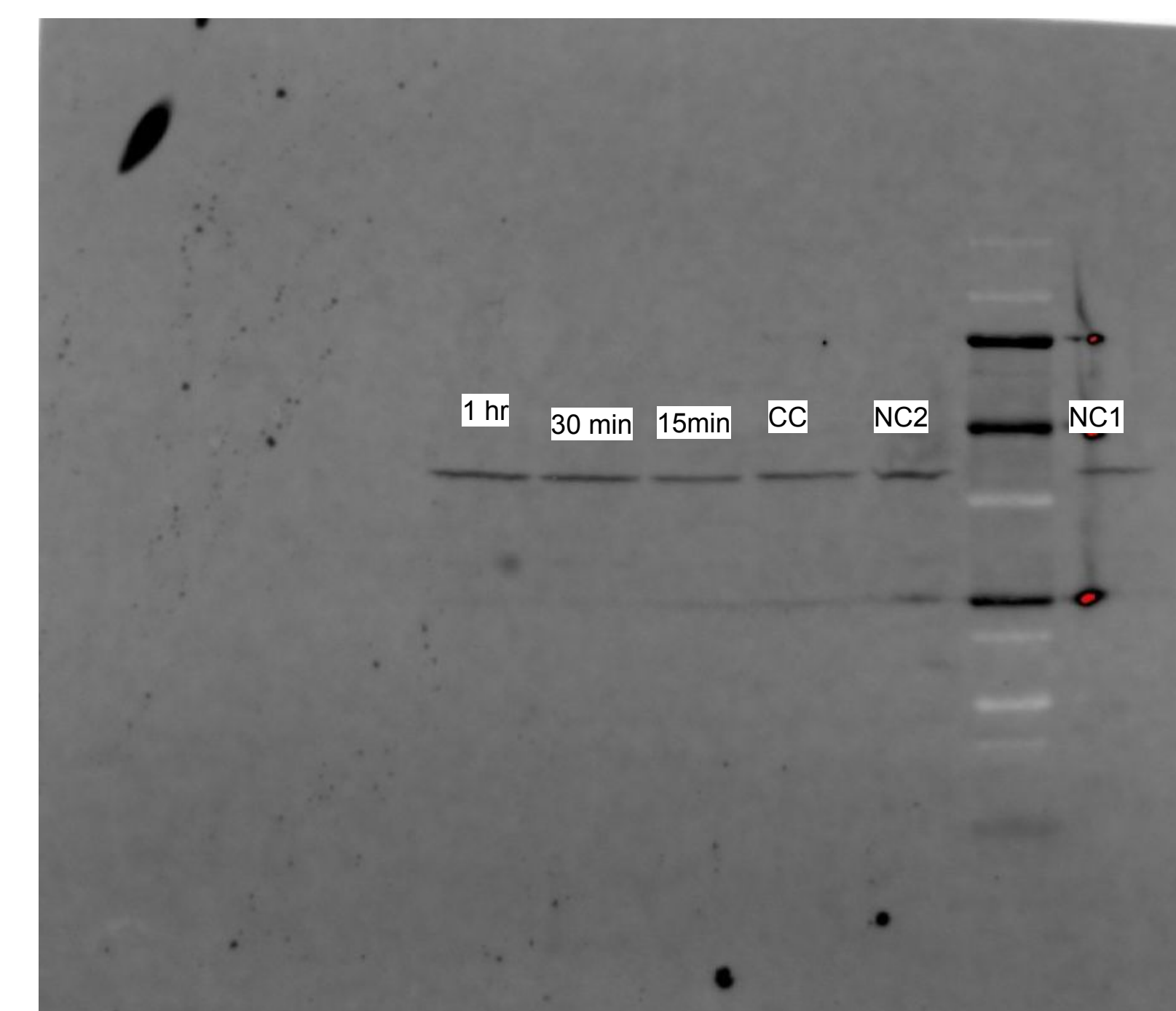


Figure 5: Image of anti-actin western blot. Image taken by researcher using Bio-Rad Chemidoc.

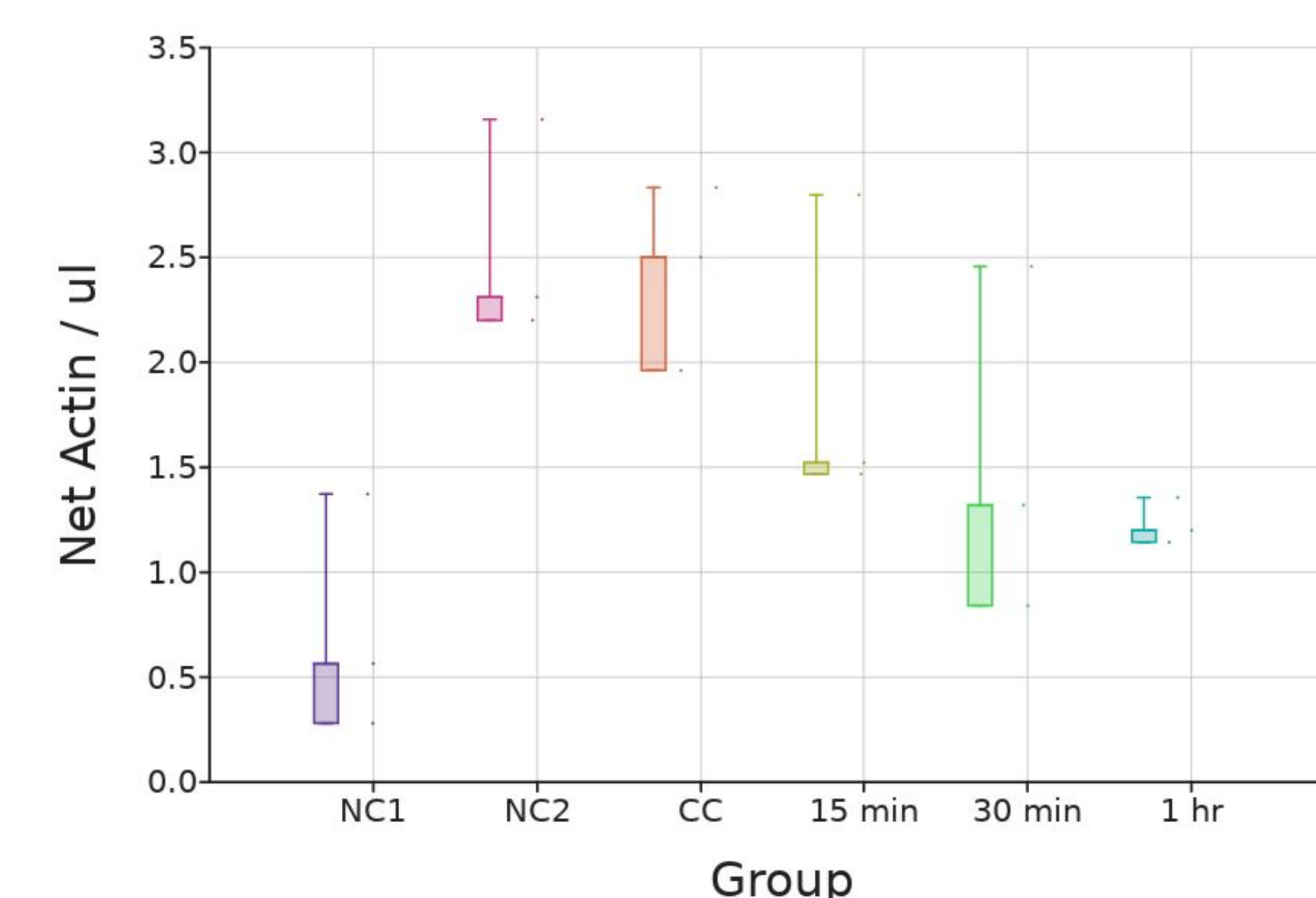


Figure 6: Expression of actin in hearts of HCM flies treated with exercise is reduced compared to the non-treated controls. n=30 flies per group. Graph created by researcher in DataClassroom.

Results

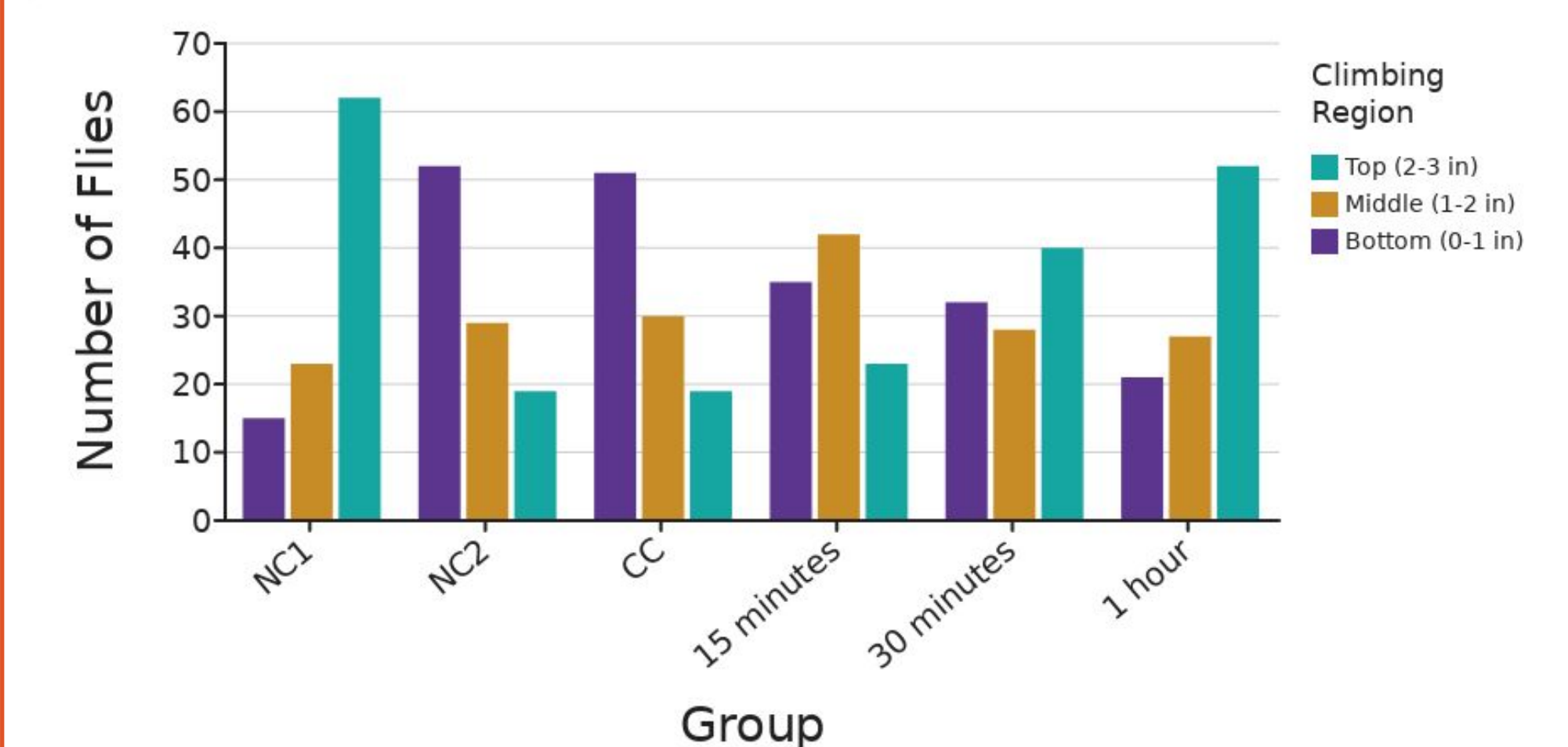


Figure 7: Exercise improves negative geotaxis in HCM flies. n=100 flies per group. Graph created by researcher in DataClassroom.

Data Set	Pair	P-Value	KW P-Value
Western Blot	NC1 - NC2	0.02	
Western Blot	NC1 - 15 min	0.19	
Western Blot	NC1 - 30 min	0.56	
Western Blot	NC1 - 1 hr	0.94	
Western Blot	NC2 - 15 min	0.73	
Western Blot	NC2 - 30 min	0.30	
Western Blot	NC2 - 1 hr	0.09	
Bottom (0-1 in)	NC1 - NC2	<0.01	0.01
Bottom (0-1 in)	NC1 - 15 min	0.06	
Bottom (0-1 in)	NC1 - 30 min	0.21	
Bottom (0-1 in)	NC1 - 1 hr	1.00	
Bottom (0-1 in)	NC2 - 15 min	0.36	
Bottom (0-1 in)	NC2 - 30 min	0.11	
Bottom (0-1 in)	NC2 - 1 hr	<0.01	<0.01
Top (2-3 in)	NC1 - NC2	<0.01	
Top (2-3 in)	NC1 - 15 min	<0.01	
Top (2-3 in)	NC1 - 30 min	0.52	
Top (2-3 in)	NC1 - 1 hr	1.00	
Top (2-3 in)	NC2 - 15 min	1.00	
Top (2-3 in)	NC2 - 30 min	0.21	
Top (2-3 in)	NC2 - 1 hr	<0.01	<0.01

Table 2: Statistics for the western blot and climbing assay. Table created by researcher using Google Sheets.

Conclusions

- Hypertrophic cardiomyopathy model was successfully established** as measured by an increase in actin expression in the heart and a reduction in negative geotaxis.
- Expression of actin** as a measure of increased heart size is **not significantly reduced** in flies that were treated with 1 hour of low-intensity exercise ($p = 0.09$).
- Negative geotaxis in HCM flies improved** with 60 minutes of low-intensity exercise ($p < 0.01$).
- Actin expression and negative geotaxis were both generally improved** in HCM flies treated with low-intensity exercise.
- The negative geotaxis data provides support for the use of low-intensity exercise as a viable treatment option for patients with HCM.
- Although expression of actin was not significantly reduced there was a trend, and future work will focus on refining these results with additional data.

References

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